for them as well. Figure 3 shows nine others, namely Beta (with alpha = beta = 0.5), Cauchy, Chi-Square, Exponential, Gamma, Laplace, Log Normal, Poisson and Uniform, using their respective functions, as shown in the code below:

```
octave:1> B=-20:0.5:20;
octave:2> subplot(3, 3, 1);
octave:3> hist(40*(betarnd(0.5, 0.5, 100000, 1) - 0.5), B);
octave:4> title("Beta");
octave:5> subplot(3, 3, 2);
octave:6> hist(cauchy_rnd(0, 1, 100000, 1), B);
octave:7> title("Cauchy");
octave:8> subplot(3, 3, 3);
octave:9> hist(chi2rnd(5, 100000, 1), B);
octave:10> title("Chi Square");
octave:11> subplot(3, 3, 4);
octave:12> hist(exprnd(5, 100000, 1), B);
octave:13> title("Exponential");
octave:14> subplot(3, 3, 5);
octave:15> hist(gamrnd(5, 1, 100000, 1), B);
octave:16> title("Gamma");
octave:17> subplot(3, 3, 6);
octave:18> hist(laplace_rnd(100000, 1), B);
octave:19> title("Laplace");
octave:20> subplot(3, 3, 7);
octave:21> hist(lognrnd(0, 1, 100000, 1), B);
octave:22> title("Log Normal");
octave:23> subplot(3, 3, 8);
octave:24> hist(poissrnd(5, 100000, 1), B);
octave:25> title("Poisson");
octave:26> subplot(3, 3, 9);
octave:27> hist(unifrnd(-10, 10, 100000, 1), B);
octave:28> title("Uniform");
octave:29>
```

Miscellaneous helpers

Apart from the standard ones already discussed, there are quite a few miscellaneous useful functions provided by Octave. Here are a few:

perms() - Generates the n! permutations of the data points

values() - Sorts the data into non-repeating ascending order

range() - Returns the difference between the maximum and the minimum data points

Here's a demonstration:

```
$ octave -qf
octave:1> perms([0 1 2])
ans =

   0   1   2
   1   0   2
   0   2   1
   1   2   0
   2   0   1
   2   1   0

octave:2> values([1 4 -9 -7 0 -6 12 -90])
ans =

  -90
   -9
   -7
   -6
    0
    1
    4
   12

octave:3> range([1 4 -9 -7 0 -6 12 -90])
ans =  102
octave:4>
```

What's next?

Till date, in all our mathematical explorations, we have been always dealing with numbers. Often, we come across scenarios wherein we would like a symbolic or an analytical solution, and use numbers only at the end. This is not possible with any of the OSS tools we have discussed till now. But there are other tools such as Maxima, where we are headed next.

By: Anil Kumar Pugalia

A hobbyist in open source hardware and software, with a passion for mathematics, the author is a gold medallist from the Indian Institute of Science. His exploration of mathematics in every aspect of life dates back to the 1990s. He shares his experiments with Linux and embedded systems at weekend workshops. He can be reached at *email@sarika-pugs.com* and *http://sysplay.in*.